

Mask-Aware β -VAE for IV Surface Inpainting

Progress Report

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EE5610: Term Project

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Executive Summary

Goal. Learn a compact latent model (mask-aware β -VAE) to *reconstruct + inpaint* daily IV surfaces on a fixed (k, T) grid.

Deliverables (this report).

- Data bundle (.pt) with x , mask, grids, dates
- Model + training protocol
- Baseline (bicubic)
- Result tables + panels

Planned next. One small ablation (β or TV), lightweight no-arbitrage violation counts, final curated panels.

Data & Grid

Universe. SPY options (2019–2024).

Grid. 64×64 , $k = \ln(K/S) \in [-0.5, 0.5]$,
 $T \in [7, 365]$ days.

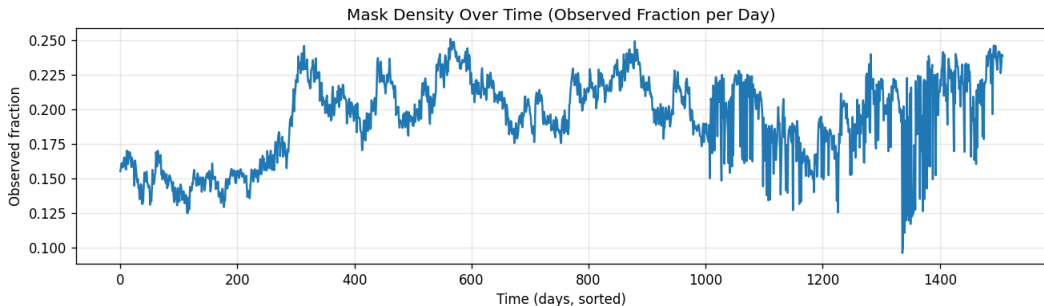
Per day. Field $x \in \mathbb{R}^{64 \times 64}$, mask $m \in \{0, 1\}^{64 \times 64}$.

Norm. Global min-max on observed cells.

Split. Time-ordered 70/15/15.

Dataset stats

Days total	1508
Train / Val / Test	1055 / 226 / 227
Mean obs. frac.	0.193
Min/Max obs. frac.	0.096 / 0.251
IV range (obs)	[0.0115, 1.000]



Bicubic Baseline: Pipeline

Bicubic interpolation on (k, T) :

- Fit on observed cells ($m=1$)
- Eval on mask; report masked-MAE
- Fast; classical yardstick

Data

- $\mathbf{x}[N, 1, H, W]$,
 $\mathbf{mask}[N, 1, H, W]$
- Grids: $T[H]$, $k[W]$, dates
[N]
- Split: train/val/test
- Pick test day i

Grid (Obs. subset)

Mode: *Recon/Inpaint*

- Recon: $use_mask = mask[i]$
- Inpaint: $keep \sim \text{Bern}(p) \odot mask[i]$
- Scatter $P = \{(k_j, T_u, \sigma_{uj}) \mid use_mask[u, j] = 1\}$

Model: Bicubic interp

`griddata(..., 'cubic')`

Fallbacks: 'linear' $\rightarrow$ 'nearest'

Output: $\hat{\sigma} \in \mathbb{R}^{H \times W}$

Model: Mask-Aware β -VAE

Input (2-channels): $[x \odot m_{\text{in}}, m_{\text{in}}]$, where m_{in} is the full mask for reconstruction or a thinned mask for inpainting.

Encoder: Four convolutional layers with strides of 2 and ReLU activations progressively downsample the input from 64×64 to 4×4 , followed by flattening and two dense layers producing μ and $\log \sigma^2$ for a latent vector $z \in \mathbb{R}^{16-32}$.

Latent Sampling: $z = \mu + \epsilon \odot \exp(0.5 \log \sigma^2)$, with $\epsilon \sim \mathcal{N}(0, I)$.

Decoder: A dense layer reshapes z into $4 \times 4 \times 128$, followed by four transposed convolutional layers with ReLU activations that upsample back to 64×64 , ending with a sigmoid output layer producing $\hat{x}$.

Loss: Masked reconstruction loss (MSE) + β -weighted KL divergence + smoothness and arbitrage regularizers.

Objective (Masked ELBO + Soft Priors)

$$\mathcal{L} = \underbrace{\frac{\sum (m^{(\text{loss})} \odot (x - \hat{x})^2)}{\sum m^{(\text{loss})}}}_{\text{masked-MSE}} + \beta \text{KL}(q_\phi(z|x) \parallel \mathcal{N}(0, I)) + \lambda_{\text{TV}} \text{TV}(\hat{x}) + \lambda_{\text{cal}} \text{CalMono}(\hat{x}).$$

$$\text{TV}(\hat{x}) \approx \sum_{i,j} \sqrt{(\hat{x}_{i+1,j} - \hat{x}_{i,j})^2 + (\hat{x}_{i,j+1} - \hat{x}_{i,j})^2 + \varepsilon}, \quad \text{CalMono}(\hat{x}) = \sum_{i,j} \max\{0, \hat{\sigma}_{i+1,j}^2 T_{i+1} - \hat{\sigma}_{i,j}^2 T_i\}.$$

What each symbol means & why it's there

- **Data & prediction:** $x \in [0, 1]^{H \times W}$ is the normalized IV surface on a fixed grid; $\hat{x}$ is the decoder's reconstruction.
- **Masking:** $m^{(\text{loss})} \in \{0, 1\}^{H \times W}$ selects the cells used in the loss (observed cells for reconstruction; the hidden subset for inpainting). $\odot$ is elementwise product.
- **Latent model:** $q_\phi(z|x)$ is the encoder's posterior over latent z ; the prior is $\mathcal{N}(0, I)$. The β -scaled KL term regularizes z (prevents overfit, encourages smoother latents). Typical $\beta \in [1, 2]$.
- **TV smoothness:** $\text{TV}(\hat{x})$ penalizes large local gradients across the (T, k) grid (indices i for maturity rows, j for moneyness columns). Small λ_{TV} denoises without washing out structure; ε prevents division by zero.

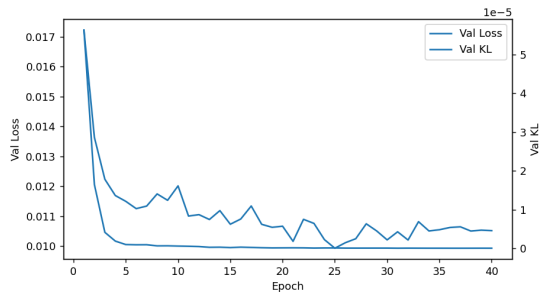
Training Protocol (Implemented)

Optimizer. Adam (lr 10^{-3} , cosine decay), batch 64, ~ 40 epochs.

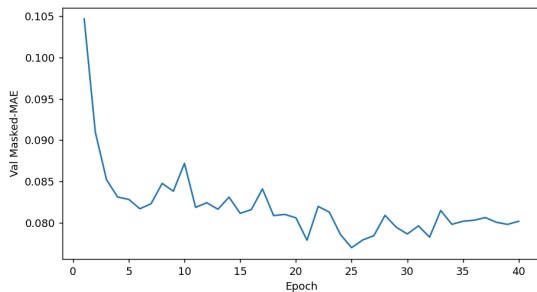
Curriculum. Warmup (full recon) $\rightarrow$ inpainting (randomly hide observed cells).

Current. $\beta=2$, $\lambda_{TV}=5 \times 10^{-5}$, $\lambda_{arb}=5 \times 10^{-3}$.

Training curves (loss/KL) [Val]



Masked-MAE vs epoch [Val]



Mask Curriculum (Pseudo-code)

```
for epoch in range(E):
    if epoch < warmup:
        m_in, m_loss = m, m
    else:
        keep = Bernoulli(p).sample(m.shape) # p ~ U(0.2, 0.8)
        m_in = m * keep # hide some observed
        m_loss = m * (1 - keep) # eval on hidden
        observed
        if m_loss.sum() < min_obs: m_loss = m

    x_in = concat([x * m_in, m_in], dim=1)
    x_hat, mu, logvar = VAE(x_in)
    L = maskedMSE(x, x_hat, m_loss) + beta*KL(mu, logvar)
        + l_tv*TV(x_hat) + l_cal*CalMono(x_hat)
    step(L)
```

Quantitative Results

Table 1. Reconstruction (observed cells only)

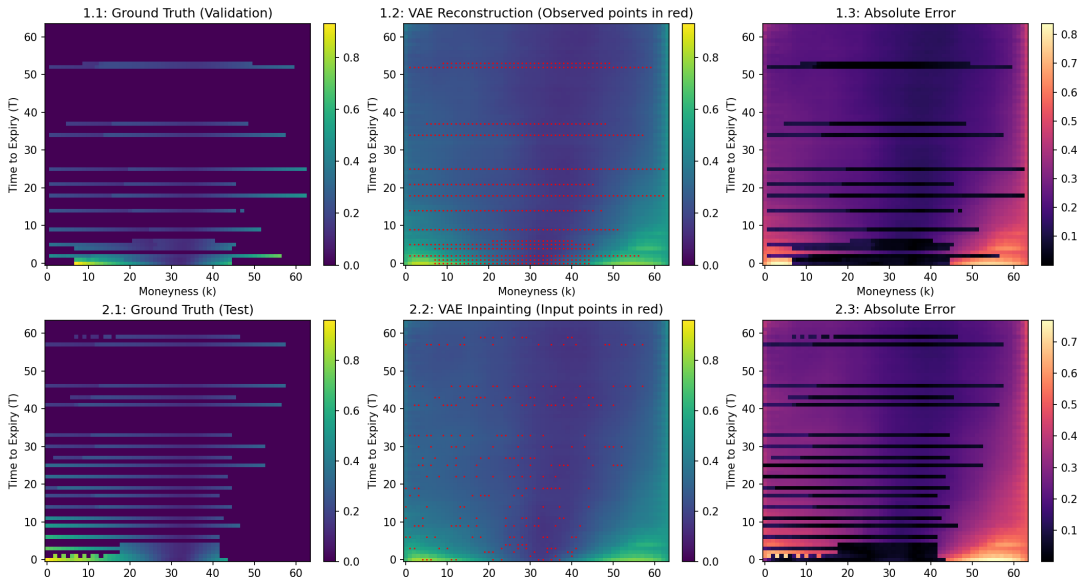
Model	Split	Masked-MAE	MSE	Notes
Bicubic (baseline)	Val	0.000	0.000	interpolation
β -VAE ($\beta=2$)	Val	0.0777463	0.010020	warmup/best
β -VAE ($\beta=2$)	Test	0.075663	0.010025	best ckpt

Table 2. Inpainting (held-out observed cells)

Model	Split	Masked-MAE	MSE	Notes
Bicubic (baseline)	Val	0.010020	0.000691	on held-out
β -VAE ($\beta=2$)	Val	0.077107	0.009935	on held-out

Qualitative Performance

Qualitative VAE Performance



Next Steps & Future Prospects

Immediate To-Do:

- Fine-tune hyperparameters and regularizers (β , λ_{TV} , λ_{arb}).
- Improve mask curriculum and add structured missing patterns.
- Add validation metrics (Masked-MAE/MSE, arbitrage violation counts).

Model Tweaks (Performance Gains):

- Larger latent space, residual decoder blocks.
- CoordConv inputs and adaptive regularization.
- Explore per-cell z-score normalization.

Future Extensions:

- **CVAE**: Condition on market regime or volatility level.
- **DDPM**: Use diffusion models for stochastic surface generation.
- Uncertainty estimation & cross-asset generalization.

Note: Further improvements (CVAE/DDPM) may enhance realism but require longer training times.